

Suraj Kumar

PhD, IIT Indore

✉ surajphdiit26@gmail.com

🌐 Google Scholar

🌐 LinkedIn



Professional Experience

- 01/2025 – 06/2025 **Visiting Scholar.** Collaborated with Dr. Chandranath Adak at IIT Patna, on NLP projects focusing book genre prediction and hotel complaint analysis using user reviews.
- 07/2024 – 12/2024 **Research Intern.** Worked with Dr. Arani Bhattacharya at IIIT Delhi, to benchmark real-time gRPC-based ML services for Autonomous Vehicles (AVs).
- 09/2021 – 06/2023 **Junior Research Fellow.** Collaborated with Dr. Soumi Chattopadhyay at IIIT Guwahati, developed graph-based learning models for real-time web service recommendation.
- 01/2020 – 08/2021 **Assistant Professor.** Dept. of CSE, GLA University, Mathura. Delivered courses in AI & ML, supervised student research, and guided projects on practical applications.
- 06/2018 – 08/2018 **Research Intern.** Worked with Dr. Aditya Nigam at IIT Mandi. Proposed and developed a video steganography framework using 3D-CNN and Hourglass architectures.

Education

- 2023 – 2026, CPI: 9.4/10 **Ph.D., CSE Dept., IIT Indore**
Title: *Anomaly-Resilient and Robust Learning Frameworks for Real-time QoS Prediction.*
Supervisor: Dr. Soumi Chattopadhyay
- 2017 – 2019, CPI: 9.2/10 **M.Tech., CSE Dept., AMU Aligarh.**
Title: *Video Steganography using Spatio-Temporal features.*
Supervisor: Prof. Saiful Islam
- 2013 – 2017, CPI: 8.1/10 **B.Tech., CSE Dept., AMU Aligarh**
Title: *Vehicle Tracking and Automation System.*
Supervisor: Mr. MA Qadeer

Research Publications

Journal Articles

- 1 **S. Kumar**, S. Chattopadhyay, and C. Adak, "Anomaly Resilient Temporal QoS Prediction using Hypergraph Convolved Transformer Network," *IEEE Transactions on Network and Service Management*, pp. 1–1, 2026. DOI: 10.1109/TNSM.2026.3674650.
- 2 U. K. Nareti, S. Chattopadhyay, C. Adak, P. Mallick, **S. Kumar**, A. V. Daga, A. Wase, and A. Roy, "Adaptive Data-Resilient Multi-Modal Hierarchical Multi-Label Book Genre Identification," *IEEE Transactions on Multimedia*, pp. 1–1, 2026, [Accepted]. URL: <https://arxiv.org/abs/2505.03839>.
- 3 **S. Kumar** and S. Chattopadhyay, "ARRQP: Anomaly Resilient Real-time QoS Prediction Framework with Graph Convolution," *IEEE Transactions on Services Computing*, no. 01, pp. 1–16, Apr. 2025, ISSN: 1939-1374. DOI: 10.1109/TSC.2025.3565376.
- 4 **S. Kumar**, S. Chattopadhyay, and C. Adak, "TPMCF: Temporal QoS Prediction Using Multi-Source Collaborative Features," *IEEE Transactions on Network and Service Management*, vol. 21, no. 4, pp. 3945–3955, Aug. 2024, ISSN: 1932-4537. DOI: 10.1109/TNSM.2024.3395428.

Conference Proceedings

- 1 K. Priya, **S. Kumar**, et al., "Graph Convolutional Teacher-Student Framework for Writer Inspection from Handwritten Words with Intra-Writer Variability," in *2025 19th IAPR International Conference on Document Analysis and Recognition (ICDAR)*, Springer, 2025.
- 2 S. Chattopadhyay, R. Chanda, **S. Kumar**, and C. Adak, "OffDQ: An Offline Deep Learning Framework for QoS Prediction," in *Proceedings of the ACM Web Conference 2022*, ser. WWW '22, Virtual Event, Lyon, France: Association for Computing Machinery, 2022, pp. 1987–1996, ISBN: 9781450390965. DOI: 10.1145/3485447.3512107.

- 3 **S. Kumar** and S. Chattopadhyay, "TRQP: Trust-Aware Real-Time QoS Prediction Framework Using Graph-Based Learning," in *Service-Oriented Computing: 20th International Conference, ICSOC 2022, Seville, Spain, November 29 – December 2, 2022, Proceedings*, Seville, Spain: Springer-Verlag, 2022, pp. 143–152, ISBN: 978-3-031-20983-3.  DOI: 10.1007/978-3-031-20984-0_10.
- 4 A. Jaiswal, **S. Kumar**, and A. Nigam, "En-VStegNET: Video Steganography using spatio-temporal feature enhancement with 3D-CNN and Hourglass," in *2020 International joint conference on neural networks (IJCNN)*, IEEE, 2020, pp. 1–8.  DOI: 10.1109/IJCNN48605.2020.9206921.
- 5 A. Mishra*, **S. Kumar***, A. Nigam, and S. Islam, "VStegNET: Video Steganography Network using Spatio-Temporal features and Micro-Bottleneck," in *30th British Machine Vision Conference 2019, BMVC 2019, Cardiff, UK, September 9-12, 2019, * Joint First Author*, BMVA Press, 2019, p. 274.  URL: <https://bmvc2019.org/wp-content/uploads/papers/0966-paper.pdf>.

Pre-Prints

- 1 R. Bukke, S. Pandey, **S. Kumar**, S. Chattopadhyay, and C. Adak, *Agentic Multi-Persona Framework for Evidence-Aware Fake News Detection*, 2025. arXiv: 2512.21039 [cs.IR].  URL: <https://arxiv.org/abs/2512.21039>.
- 2 **S. Kumar**, A. Kumar, and S. Chattopadhyay, *SHARP-QoS: Sparsely-gated Hierarchical Adaptive Routing for joint Prediction of QoS*, 2025. arXiv: 2512.17262 [cs.LG].  URL: <https://arxiv.org/abs/2512.17262>.
- 3 **S. Kumar**, U. K. Nareti, S. Chattopadhyay, C. Adak, and P. Mallick, *Blurb-Refined Inference from Crowdsourced Book Reviews using Hierarchical Genre Mining with Dual-Path Graph Convolutions*, 2025. arXiv: 2512.21076 [cs.IR].  URL: <https://arxiv.org/abs/2512.21076>.
- 4 U. K. Nareti, **S. Kumar**, S. Pandey, S. Chattopadhyay, and C. Adak, *ProtoSiTex: Learning Semi-Interpretable Prototypes for Multi-label Text Classification*, 2025. arXiv: 2510.12534 [cs.AI].  URL: <https://arxiv.org/abs/2510.12534>.

Skills

Languages	 Strong reading, writing and speaking competencies for English, and Hindi.
Coding	 Python, PyTorch, TensorFlow
Misc.	 Academic research, teaching, $\text{\LaTeX}$ typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

-  **General Secretary** in Student Gymkhana Council, IIIT Guwahati, India (2022-2023).
-  **Student Volunteer** in multiple Workshops at CSE Dept., AMU Aligarh and in IWADL at IIT Mandi (2018-2019).

Certification

-  **Exam Qualified.** GATE (2020), GATE (2017), and UGC-NET (2019).

Academic Services

-  **Reviewer.** IEEE Trans. on Network and Service Management (2026), IEEE Internet of Things Journal (2026), IEEE Trans. on Cloud Computing (2025), ACM Trans. on Internet Technology (2025), Computer Network (Elsevier, 2025), Future Generation Computer Systems (Elsevier, 2025), IEEE Trans. on Multimedia (2021–22)

References

Dr. Soumi Chattopadhyay
Assistant Professor
IIT Indore, India
 soumi@iiti.ac.in

Dr. Chandranath Adak
Assistant Professor
IIT Patna, India
 chandranath@iitp.ac.in

Dr. Arani Bhattacharya
Assistant Professor
IIIT Delhi, India
 arani@iiitd.ac.in